

A Project report on
ELECTRONIC VOTING SYSTEM USING
BLOCKCHAIN TECHNOLOGY

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY
In
COMPUTER SCIENCE AND ENGINEERING (IOT & CYBERSECURITY
INCLUDING BLOCKCHAIN TECHNOLOGY)

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[2025 – 2026]

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BONAFIDE CERTIFICATE

This is to certify that the project work entitled **ELECTRONIC VOTING SYSTEM USING BLOCKCHAIN TECHNOLOGY** is the bonafide work of **IVATURI N S V R KASHYAP (22B91A4721), SUREDDY B D S N RISHI (23B95A4712), PENMETSA RAKSHITHA (22B91A4737), DEVARAKONDA PAVAN KALYAN (23B95A4705)** who carried out the project work under my supervision in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING.

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SELF DECLARATION

We hereby declare that the project work entitled **ELECTRONIC VOTING SYSTEM USING BLOCKCHAIN TECHNOLOGY** is a genuine work carried out by us in B.Tech., Computer Science and Engineering(IoT, Cybersecurity Including Blockchain Technology) at SRKR Engineering College(A), Bhimavaram and has not been submitted either in part or full for the award of any other degree or diploma in any other institute or University.

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ABSTRACT

Electronic voting systems have emerged as an efficient alternative to traditional paper-based voting due to their speed, convenience, and ease of result computation. However, many existing e-voting systems rely on centralized architectures, where voting data is stored and controlled by a single authority. This centralized structure makes such systems vulnerable to cyber-attacks, insider manipulation, data tampering, and single points of failure. Additionally, limited transparency and lack of strong identity verification mechanisms reduce trust in electronic election processes.

To address these challenges, this project proposes a Blockchain-Based Online Voting System with Biometric Face Verification that leverages the decentralized and immutable properties of blockchain technology. In the proposed system, voter registration and facial enrollment are performed off-chain by the administrator, while vote recording is carried out on the blockchain through smart contracts. At the polling stage, voter identity is verified using real-time facial authentication, ensuring that only authorized and registered voters can cast their votes. A simple and user-friendly web interface is provided for both administrative and polling operations, requiring no technical knowledge from voters.

Once authentication is successful, votes are securely transmitted to the blockchain via smart contracts, which enforce voting rules such as preventing duplicate voting and validating candidate selection. Each vote is stored as an immutable transaction on the blockchain ledger, ensuring transparency and tamper resistance. The system maintains voter anonymity while guaranteeing vote integrity and secure storage. This solution is particularly suitable for small to medium-scale elections such as college or organizational voting, demonstrating the practical implementation of blockchain and biometric authentication in building a secure and trustworthy electronic voting system.

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LIST OF SYMBOLS

- API – Application Programming Interface
- UI – User Interface
- IoT – Internet of Things
- HTML – HyperText Markup Language
- CSS – Cascading Style Sheets
- JS – JavaScript
- JSON – JavaScript Object Notation
- HTTP – Hypertext Transfer Protocol
- URL – Uniform Resource Locator
- ID – Identifier
- DB – Database
- SQL – Structured Query Language
- npm – Node Package Manager
- REST – Representational State Transfer
- P2P – Peer to Peer Network
- DLT – Distributed Ledger Technology
- EVM – Electronic Voting Machine
- ETH – Ethereum
- TxID – Transaction Identifier
- OS – Operating System
- CPU – Central Processing Unit
- PoW/PoS – Proof of work / Proof of Stake

1. INTRODUCTION

Blockchain is a decentralized and distributed digital ledger technology that allows data to be recorded and shared securely across a network of computers. It was first introduced in 2008 by an anonymous person or group known as Satoshi Nakamoto as the foundation for Bitcoin, enabling peer-to-peer digital payments without the need for a central authority. Initially, blockchain was used only for cryptocurrency transactions, but with the launch of Ethereum in 2015, it expanded to include smart contracts—self-executing programs that automatically carry out agreements. Today, blockchain has applications far beyond digital currency, including finance, healthcare, supply chain management, digital identity, and electronic voting, thanks to its key features like decentralization, transparency, immutability, security, and distributed data storage.

Electronic Voting Machines (EVMs) are digital devices designed to record and count votes efficiently, replacing traditional paper-based voting systems. They aim to simplify the voting process, minimize human error, and deliver faster election results. However, despite these advantages, EVMs face several challenges. Voters cannot independently verify whether their vote was recorded correctly, and election data is often stored in centralized systems, making it vulnerable to hacking, technical failures, or manipulation. Additionally, the limited ability to audit votes and concerns over security can reduce public confidence in the fairness and accuracy of election results.

The proposed Blockchain-Based Online Voting System with Biometric Face Verification integrates blockchain technology with facial authentication to create a secure and trustworthy voting environment. In this system, voter registration and face enrollment are handled off-chain, while the actual vote recording is performed on the blockchain through smart contracts. Real-time face verification ensures that only legitimate voters can cast their votes, thereby preventing impersonation and duplicate voting. By combining biometric authentication with immutable blockchain storage, the system enhances election integrity, transparency, and security, making it suitable for small to medium-scale elections such as colleges and organizational institutions.

2. LITERATURE REVIEW

A detailed literature review was conducted to study existing research related to electronic voting systems and the application of blockchain technology in e-voting. The objective of this review is to analyze existing solutions, understand how blockchain improves security and transparency, and identify limitations and research gaps. A total of 10 research papers from reputed publishers such as IEEE, ACM, Springer, MDPI, Elsevier, ITM Conferences, IJST, and ResearchGate were studied.

Table 2.1 : Literature S

Paper Title	Authors	Publisher & Year	What It Says (Key Points)	Limitations Identified
A Comprehensive Analysis of Blockchain-Based Voting Systems: Enhancing Transparency and Security	Md. Atik Ullah Khan Atik, Sumaiya Sultana Tarin, Mushfequr Rahman, Ahsanul Azmain Alvi, Md. Faruk Abdullah Al Sohan	ACM Conference (ICCA), 2024	• Blockchain improves transparency and security in voting systems • Decentralization removes centralized manipulation risks • Cryptographic methods protect vote integrity and voter privacy	• Mostly analytical and survey-based • Lacks a simple end-to-end implementation with live counting
Analysis of Blockchain Solutions for E-Voting: A Systematic Literature Review	Ali Benabdallah, Antoine Audras, Louis Coudert, Mohamad Badra, Nour El Madhoun	IEEE Access, 2022	• Blockchain removes trusted third party • Smart contracts automate voting processes • Improves transparency and resistance to fraud	• Focuses on review rather than implementation • Scalability and privacy challenges remain open
Blockchain-Powered E-Voting: A Novel Approach to Secure Voter Authentication	B Sujatha, Yarrampati Ganesh, N Leelavathy, R Tamilkodi, Bayye Sandhya	Indian Journal of Science and Technology, 2024	• Uses Ethereum and smart contracts • Integrates IPFS for decentralized storage • Enhances voter authentication and election automation	• Uses advanced biometrics increasing system complexity • High infrastructure requirements

Blockchain-Based Voting Systems: Enhancing Transparency and Security in Electoral Processes	Dokku Durga Bhavani, Gayathri R, Bhagavanthu M, Adlin Sheeba, Bhuvaneshwari P Maria Sampooram M	ITM Web of Conferences, 2025	• Blockchain ensures immutability of votes • Prevents vote manipulation using decentralized ledgers • Improves transparency and voter confidence	• Scalability issues for large-scale elections • Requires high digital literacy
Blockchain for Electronic Voting System — Review and Open Research Challenges	Uzma Jafar, Mohd Juzaidin Ab Aziz, Zarina Shukur	MDPI Sensors, 2021	• Blockchain offers decentralization and end-to-end verifiability • Improves trust and auditability in elections	• Privacy protection and transaction speed not fully solved
Blockchain-Based E-Voting Mechanisms: A Survey and Proposal	Matthew Sharp, Laurent Njilla, Chin-Tser Huang, Tieming Geng	MDPI Network, 2024	• Reviews architectures and vote counting methods • Proposes improved counting techniques • Shows feasibility on constrained devices	• Complex design unsuitable for simple elections • Not focused on user usability
Blockchain-Based Electronic Voting Systems: A Case Study in Morocco	Tarik Chafiq, Rida Azmi, Ouadoud Mohammed	ScienceDirect (Elsevier), 2024	• Real-world case study using blockchain • Reduces fraud and manipulation • Enhances transparency and accuracy	• Designed for specific blockchain platforms (Solana) • Limited generalization
Blockchain for Securing Electronic Voting Systems: A Survey of	Henry O. Ohize, Adeiza James Onumanyi, Buhari U. Umar,	Springer (Cluster Computing), 2025	• Blockchain improves security, transparency & auditability • Surveys global implementations	• High resource consumption • Infrastructure and scalability challenges

Architectures, Trends, Solutions, and Challenges	Lukman A. Ajao, Rabi O. Isah, Eustace M.		(Estonia, Switzerland) • Highlights cryptographic measures	
E-Voting Using Blockchain Technology	Abhishek Subhash Yadav, Yash Vandesh Urade, Ashish Uttamrao Thombare, Abhijeet Anil Patil	IEEE Conference / ResearchGate, 2020	• Proposes a blockchain-based e-voting framework • Uses Ethereum smart contracts to store votes	• Does not support real-time live vote counting • Focuses more on backend, limited UI design
Blockchain Based Decentralized E-Voting System: A Survey	Shashank Singh, Shivik Bansal, Sidhant Semwal	ICCS Conference (IEEE-related), 2023	• Surveys decentralized blockchain-based e-voting systems and their architectures. • Highlights key benefits such as transparency, security, and tamper resistance	• Primarily a survey with no full system implementation • Does not include real-time live vote counting. • Does not include real-time live vote counting

Summary

Paper [1] by Md. Atik Ullah Khan et al. presents a comprehensive analysis of blockchain-based voting systems that aim to enhance transparency and security in electoral processes. The study explains how blockchain decentralization eliminates centralized manipulation risks and protects vote integrity using cryptographic mechanisms. However, the work mainly focuses on analytical and survey-based discussions and lacks a complete end-to-end implementation with real-time vote counting.

Paper [2] by Ali Benabdallah et al. provides a systematic literature review of blockchain solutions for electronic voting systems. The study highlights that blockchain technology removes the need for trusted third parties and improves transparency using smart contract-based voting processes. However, the research mainly focuses on reviewing existing solutions rather than implementing a practical system, leaving scalability and privacy challenges unresolved.

Paper [3] by B. Sujatha et al. proposes a blockchain-powered voting framework that integrates Ethereum smart contracts and IPFS for decentralized data storage. The system enhances voter authentication and improves election automation through blockchain mechanisms. However, the proposed approach requires advanced biometric systems and high infrastructure resources, increasing the complexity of real-world deployment.

Paper [4] by Dokku Durga Bhavani et al. discusses blockchain-based voting systems designed to improve transparency and security in electoral processes. The study highlights that decentralized ledgers ensure vote immutability and prevent manipulation during elections. However, the proposed system faces scalability challenges for large-scale elections and requires higher digital literacy among voters.

Paper [5] by Uzma Jafar et al. reviews blockchain-based electronic voting systems and identifies key research challenges in implementing secure digital elections. The study explains how blockchain enables decentralization, trust, and verifiability in voting systems. However, issues related to privacy protection and transaction speed remain unresolved.

Paper [6] by Matthew Sharp et al. presents a survey and proposal for blockchain-based e-voting mechanisms focusing on system architectures and vote counting methods. The study demonstrates the feasibility of implementing blockchain voting systems even on constrained devices. However, the design complexity makes it unsuitable for simple elections and does not emphasize voter usability.

Paper [7] by Tarik Chafiq et al. presents a real-world case study of blockchain-based electronic voting systems implemented in Morocco. The study shows that blockchain technology can reduce election fraud and improve transparency and accuracy in voting processes. However, the system is designed for specific blockchain platforms, limiting its generalization to other environments.

Paper [8] by Henry O. Ohize et al. explores blockchain architectures, trends, and challenges in securing electronic voting systems. The study highlights global implementations of blockchain voting and emphasizes the importance of cryptographic security mechanisms. However, the system requires high computational resources and faces infrastructure and scalability challenges.

Paper [9] by Abhishek Subhash Yadav et al. proposes a blockchain-based e-voting framework using Ethereum smart contracts to securely store votes. The study demonstrates how decentralized architecture ensures tamper-proof voting records. However, the system does not support real-time vote counting and focuses mainly on backend mechanisms with limited user interface design.

Paper [10] by Shashank Singh et al. presents a survey of decentralized blockchain-based electronic voting systems and their architectures. The study highlights the advantages of transparency, security, and tamper resistance offered by blockchain technology. However, the research remains largely theoretical and does not include a complete system implementation or real-time vote counting mechanisms.

3. PROBLEM STATEMENT

With the growing use of electronic and online voting systems, ensuring security, transparency, and voter authenticity has become a significant challenge. Most existing electronic voting systems operate under centralized control, where voter data, vote storage, and result management are handled by a single authority. Such centralized architectures are highly vulnerable to cyber-attacks, insider manipulation, data tampering, and technical failures. Additionally, the absence of strong biometric verification mechanisms increases the risk of impersonation and duplicate voting, thereby reducing trust in the integrity of the electoral process.

Although blockchain technology has been proposed as a secure alternative for electronic voting, many blockchain-based voting solutions remain theoretical or partially implemented. Several systems fail to integrate strong identity verification methods, while others require complex blockchain interactions that are not user-friendly. In many cases, privacy protection, practical deployment feasibility, and ease of use are not adequately addressed. These limitations prevent widespread adoption, particularly in small to medium-scale election environments such as colleges or organizational institutions.

Therefore, there is a need for a practical, secure, and user-friendly voting system that combines decentralized blockchain technology with reliable voter authentication. The problem addressed in this project is the design and implementation of a blockchain-based online voting system integrated with biometric face verification to prevent impersonation and duplicate voting. The system aims to eliminate centralized manipulation, ensure vote immutability and transparency, protect voter anonymity, and provide a secure and trustworthy voting mechanism suitable for small to medium-scale elections

4. METHODOLOGY

4.1 EXISTING SYSTEM

The existing voting systems in many regions are based on traditional methods such as paper ballot voting or Electronic Voting Machines (EVMs). In these systems, voters are verified manually by polling officers using voter ID cards and electoral records before they are allowed to vote. Although EVMs have improved the speed and efficiency of the voting and counting process compared to manual ballot systems, the verification process still depends largely on human supervision. This manual verification may sometimes lead to issues such as identity impersonation, human errors, or unauthorized attempts to vote.

Another limitation of the existing system is that most voting infrastructures rely on centralized databases to store voter information and election data. Because of this centralized nature, the system may face risks such as data manipulation, lack of transparency, and security vulnerabilities. In addition, voters usually cannot verify whether their vote has been securely recorded after casting it. These limitations highlight the need for a more secure and transparent voting mechanism that integrates modern technologies for stronger authentication and tamper-proof vote storage

4.2 PROPOSED SYSTEM

The proposed system is a Blockchain-Based Online Voting System with Biometric Face Verification designed to provide a secure, transparent, and tamper-resistant digital voting mechanism. The system integrates blockchain technology, smart contracts, and facial biometric authentication to ensure that only eligible voters can cast their votes while maintaining anonymity and data integrity. By combining off-chain voter management with on-chain vote recording, the system eliminates centralized manipulation, prevents duplicate voting, and enhances trust in the election process. The proposed solution is suitable for controlled environments such as college, institutional, and organizational elections where security, transparency, and authenticity are critical.

System Workflow and Functionalities

1. Voter Pre-Registration and Face Enrollment

- Eligible voters are registered by the administrator in an off-chain database before the election.
- Each voter is assigned a unique voter ID and their facial biometric data is securely stored.
- The system prevents duplicate voter ID and duplicate face registrations to avoid identity misuse.

2. Officer-Based Eligibility Verification

- The polling officer enters the voter ID into the system at the booth.
- The system verifies voter existence and checks voting status.
- Only voters who have not previously voted are allowed to proceed to authentication.

3. Biometric Face Authentication

- The system activates the camera for real-time face detection.
- The captured face descriptor is compared with the stored biometric data.
- Voting access is granted only if the facial match is successful within the defined threshold.

4. Secure Vote Casting through Web Interface

- The voter selects a candidate using a simple and intuitive web-based interface.
- The interface is designed to resemble traditional electronic voting machines.
- The system confirms the selected candidate before submitting the vote.

5. Vote Recording using Smart Contracts

- The vote is transmitted to the blockchain through a smart contract.
- The smart contract enforces the one-vote-per-voter rule.
- Each vote is stored as an immutable transaction on the blockchain ledger.

6. Security and Data Integrity Mechanisms

- Blockchain ensures immutability and prevents tampering of recorded votes.
- Cryptographic techniques protect voter anonymity and transaction integrity.
- Decentralized architecture removes single points of failure and improves reliability.

4.3 REQUIREMENTS

4.3.1 SOFTWARE REQUIREMENTS

- **Operating System:**
 - Windows 10/11, Linux (Ubuntu preferred)
- **Programming Languages:**
 - JavaScript (Frontend & Backend)
 - Solidity (Smart Contracts)
- **Blockchain Platform:**
 - Ethereum-based local blockchain (Ganache)
- **Blockchain Tools:**
 - Remix IDE (Smart Contract Development & Deployment)
 - MetaMask (Wallet integration and transaction signing)
 - Ganache (local blockchain for testing)
 - Ethers.js (Blockchain interaction library)
- **Web Technologies:**
 - Frontend: HTML, CSS, JavaScript
 - Backend: Node.js with Express.js
- **Database:**
 - MySQL (Off-chain voter data and face descriptors storage)
- **Face Recognition Library:**
 - face-api.js (Biometric authentication module)
- **Integration & Communication:**
 - REST APIs for frontend-backend communication

- JSON-based data exchange
- **Development Environment:**
 - Visual Studio Code
 - Web Browser (Google Chrome recommended)

4.3.2 HARDWARE REQUIREMENTS

- **Processing Unit:**
 - CPU: Intel Core i3 or higher (Intel Core i5 recommended)
 - RAM: Minimum 4GB (8GB recommended for smooth execution)
 - GPU: Not mandatory (optional for development environments)
- **Storage:**
 - Minimum 256GB SSD or HDD
- **Network:**
 - Stable local network connection (for blockchain and server communication)
 - Internet connection optional (required only for external blockchain deployment)
- **Client Devices:**
 - Desktop systems or laptops for Admin and Polling Booth operations
 - Webcam-enabled devices for face authentication
- **Power Supply:**
 - Reliable power source to ensure uninterrupted voting operations

4.4 SYSTEM ARCHITECTURE

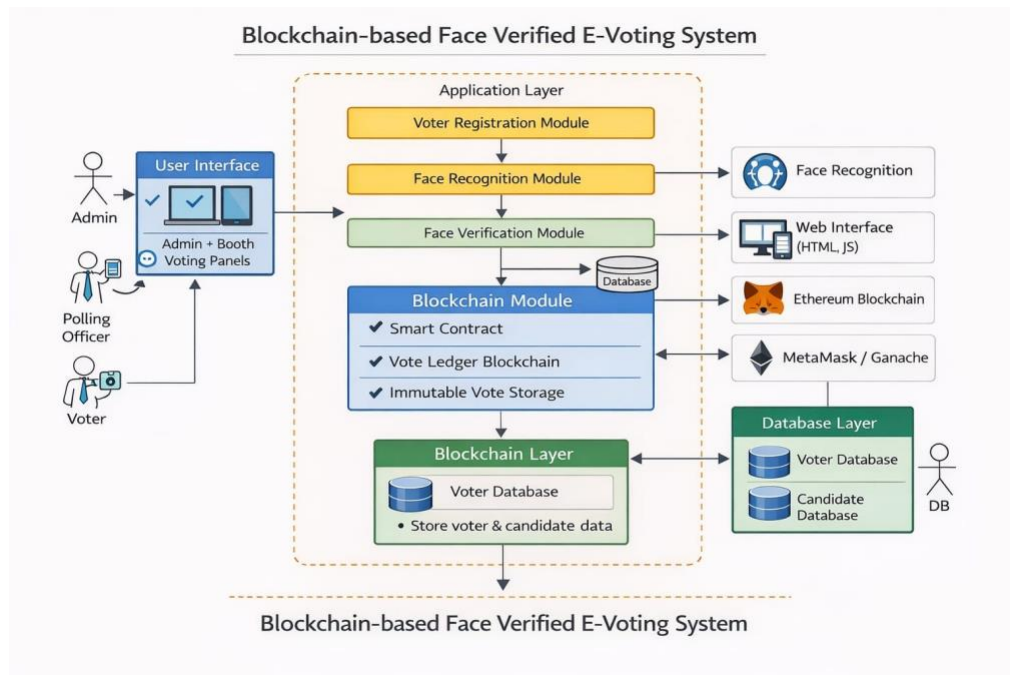


Fig. 4.1 – System Architecture

The system architecture of the proposed Blockchain-Based Online Voting System with Biometric Face Verification consists of multiple interconnected components that work together to ensure secure authentication, controlled vote casting, and tamper-proof blockchain storage. The architecture emphasizes secure interaction between the user interface, application server, biometric verification module, database, smart contracts, and the blockchain network. Each component performs a specific function while maintaining secure communication and data integrity throughout the voting process.

User Interface (Admin & Polling Booth Panels)

The User Interface serves as the primary interaction layer of the system. It includes separate panels for the administrator and the polling booth. The administrator interface allows voter registration with facial data capture and candidate management, while the polling booth interface enables voter ID verification, face authentication, and vote casting. The interface is designed to be simple, structured, and user-friendly, ensuring accessibility and ease of operation. All requests initiated through the interface are securely forwarded to the application server for processing.

Application Server

The Application Server acts as the central control unit of the system. It manages voter authentication, processes eligibility checks, validates face descriptors, and coordinates communication between the database and blockchain network. The server ensures that only authorized and verified voters are permitted to cast votes and enforces the one-vote-per-voter rule by updating voting status. After successful validation, the server interacts with the smart contract to record votes securely on the blockchain.

Face Recognition Module

The Face Recognition Module performs biometric authentication using facial descriptors. During voter registration, facial data is captured and stored securely in the database. At the polling stage, the system captures live facial input and compares it with the stored descriptor to verify identity. This module ensures that only the legitimate registered voter can proceed to vote casting, thereby preventing impersonation and identity fraud.

Database

The Database functions as an off-chain storage component for managing voter and candidate information. It securely stores voter IDs, facial descriptors, voting status flags, and candidate details. The database is responsible for checking voter eligibility and preventing duplicate voting by maintaining the hasVoted status. Sensitive blockchain transaction data is not stored here, ensuring a clear separation between identity management and immutable vote recording.

Smart Contracts

Smart contracts represent the core logic governing vote validation and recording. Once the application server submits a vote transaction, the smart contract verifies its validity and records it on the blockchain. The contract enforces voting rules such as preventing duplicate voting and ensuring that each vote corresponds to a valid candidate. Since smart contracts operate on the blockchain, their execution is transparent, automated, and tamper-resistant.

Blockchain Network

The Blockchain Network serves as a decentralized and immutable ledger for storing validated votes. Each successfully verified vote is recorded as a blockchain transaction, ensuring permanence and integrity. Due to the decentralized nature of blockchain, no single authority can modify or delete recorded votes. This enhances transparency, trust, and reliability in the election process.

Secure Communication Mechanism

Secure communication is maintained between all system components, including the user interface, application server, database, and blockchain network. Encrypted data transmission ensures that sensitive voter and transaction information is protected against unauthorized access. This end-to-end secure communication framework safeguards the integrity and confidentiality of the voting process.

4.5 DESIGN DIAGRAMS

A. USE CASE DIAGRAM

The Use Case Diagram illustrates the interactions between different actors and the Blockchain-Based Online Voting System with Face Verification. It shows how the admin manages voter registration, candidate addition, and smart contract deployment before the election begins. The polling officer verifies the voter ID and authenticates voters at the polling booth to ensure eligibility. The voter interacts with the system by providing the voter ID, performing face authentication, casting the vote, and verifying the vote using a verification code. After successful authentication, the vote is validated through smart contracts and securely stored on the blockchain, ensuring transparency, preventing manipulation, and maintaining election integrity.

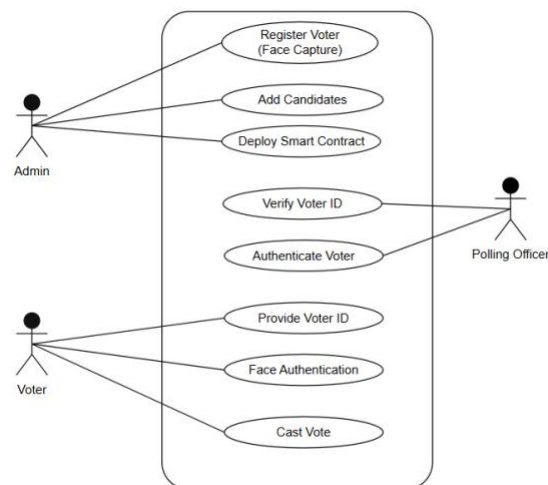


Fig. 4.2 – Use Case Diagram

Table – 4.1 – Actors Description

Actor	Use Case	Description
1. Admin	➤ Register Voter (Face Capture) ➤ Add Candidate ➤ Deploy Smart Contract	The admin prepares the election system by registering voters with their facial data, adding candidates to the voting system, and deploying the smart contract to the blockchain network.
2. Polling Officer	➤ Verify Voter ID ➤ Authenticate Voter	The polling officer verifies the voter ID and ensures that the voter is eligible before allowing them to access the voting interface.
3. Voter	➤ Provide Voter ID ➤ Face Authentication ➤ Cast Vote	The voter authenticates identity using face verification, casts a vote for the preferred candidate.

System Use Cases

1. Register Voter

This use case is performed by the admin before the election begins. The System:

- Stores voter personal details
- Captures voter facial data
- Registers the voter in the system database

2. Add Candidate

This use case allows the admin to configure the election. The System:

- Adds candidate information
- Stores candidate details in the system
- Prepares the candidate list for the voting interface

3. Deploy Smart Contract

This use case initializes the blockchain voting environment. The System:

- Deploys the smart contract to the blockchain network
- Defines voting rules such as one-vote-per-voter
- Enables secure vote storage and validation

4. Verify Voter ID

This use case is handled by the polling officer before authentication. The System:

- Validates voter identity using the voter ID
- Checks voter eligibility
- Ensures the voter has not voted previously

5. Face Authentication

This use case confirms the voter's identity. The System:

- Captures the live face using the camera
- Compares it with the registered voter face data
- Allows voting only when the face match is successful

6. Cast Vote

After successful authentication, the voter selects a candidate and casts the vote. The System:

- Records the selected candidate
- Validates the vote using the smart contract
- Prevents duplicate voting

Relationships and $\diamond$ Dependencies

- Register Voter → Face Authentication
- Verify Voter ID → Authenticate Voter
- Face Authentication → Cast Vote
- Cast Vote → Smart Contract Validation
- Smart Contract Validation → Store Vote on Blockchain
- Cast Vote → Receive Verification Code

These relationships ensure that voter authentication and validation processes are completed before storing votes on the blockchain network. This guarantees secure, transparent, and tamper-proof voting.

B. SEQUENCE DIAGRAM

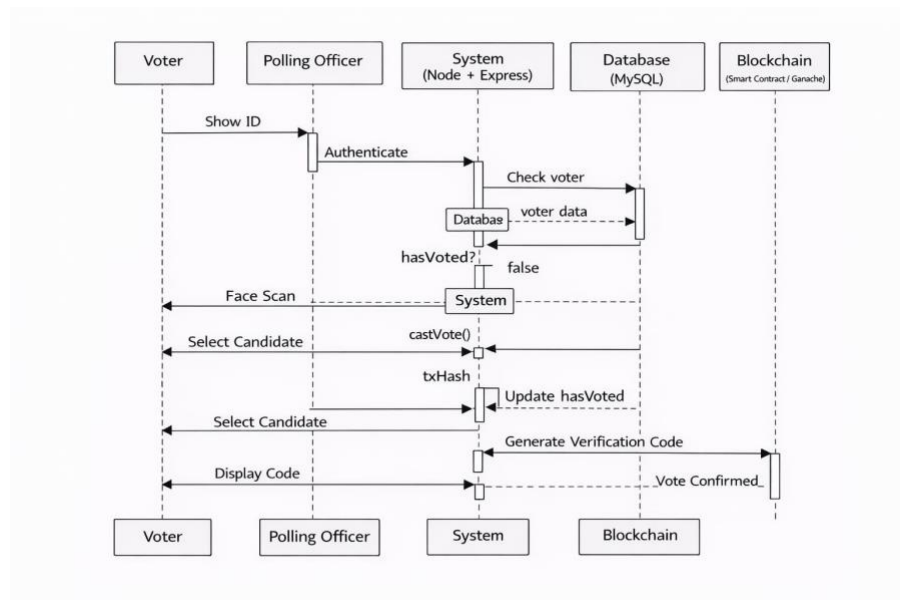


Fig. 4.3 – Sequence Diagram

Actors and Components

- Voter
- Polling Officer
- System (Node.js + Express Server)
- Database (MySQL)
- Blockchain (Smart Contract / Ganache)

Step-by-Step Interaction Flow

1. Voter Identity Verification

- The Voter shows their Voter ID to the Polling Officer.
- The Polling Officer sends an authentication request to the System.
- The System queries the Database to check:
 - Whether the voter exists
 - Whether hasVoted status is false
- The Database returns voter data.
- If hasVoted = false, the system proceeds to face authentication.

2. Face Authentication

- The System initiates a real-time face scan.
- The captured face is compared with the stored face descriptor from the database.
- If the face matches successfully, the voter is allowed to access the voting interface.

This step prevents impersonation and ensures biometric authentication.

3. Vote Casting

- The Voter selects a candidate.
- The selection is sent to the System as a castVote() request.
- The System invokes the smart contract function on the Blockchain.
- The smart contract:
 - Validates the vote
 - Enforces one-vote-per-voter rule
 - Records the vote
- The Blockchain returns a transaction hash (txHash) confirming vote submission.

4. Vote Confirmation and Database Update

- After successful blockchain confirmation, the System:
 - Updates hasVoted = true in the Database
 - Ensures the voter cannot vote again
- (If verification feature enabled) The system generates a verification code and stores it in the database.

5. Final Confirmation

- The System sends a confirmation response to the Voter.
- The interface displays:
 - Vote Successfully Recorded
 - Blockchain confirmation (optional txHash)
- This completes the secure voting transaction.

Security Enforcement in Sequence

- The diagram demonstrates that:
 - Database validation happens before vote casting.
 - Face verification happens before enabling vote submission.
 - Smart contract validation happens before permanent storage.
 - Blockchain confirmation happens before updating voter status.
- This layered validation ensures:
 - No duplicate voting
 - No impersonation
 - No centralized manipulation
 - Immutable vote storage

The Sequence Diagram clearly represents the real-time interaction between the frontend interface, backend server, database, and blockchain network throughout the voting process. It illustrates how each component communicates in a structured and time-ordered manner, beginning with voter authentication and progressing through vote validation and confirmation. The diagram emphasizes the coordinated functioning of biometric face verification, database eligibility checks, and smart contract execution before a vote is permanently recorded. By depicting these synchronized operations, it demonstrates how multiple layers of verification collectively ensure a secure, transparent, and tamper-resistant voting mechanism in the proposed Blockchain-Based Online Voting System.

C. ACTIVITY DIAGRAM

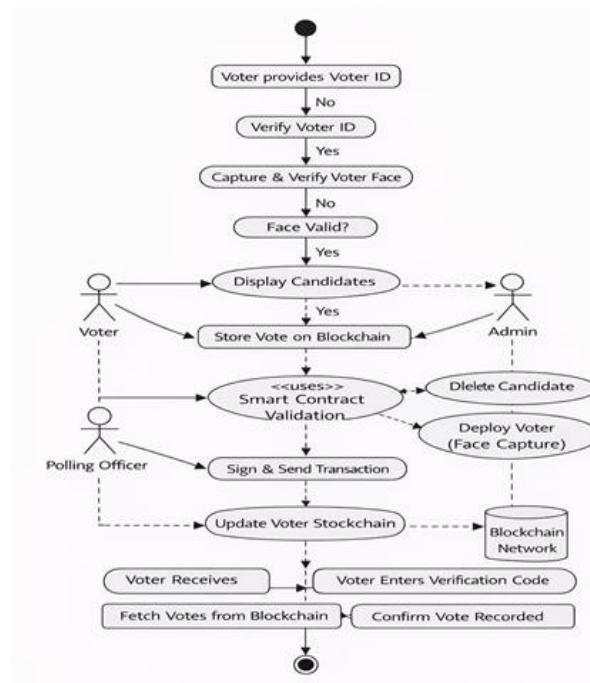


Fig . 4.4– Activity Diagram

Workflow Description

The activity flow begins when the voter arrives at the polling interface and provides their registered Voter ID. The system verifies this ID by checking the voter's eligibility and voting status in the database. If the voter does not exist or has already voted, the process terminates immediately to prevent unauthorized access or duplicate voting. If the voter is eligible, the system proceeds to the biometric authentication stage.

In the next phase, the system captures the voter's face in real time and compares it with the previously registered facial data stored in the database. A decision node evaluates whether the captured face matches the stored descriptor. If the face verification fails, the process stops to prevent impersonation. If the face is successfully validated, the system grants access to the voting interface and displays the list of available candidates.

After reviewing the candidates, the voter selects their preferred option. The selected vote is then sent to the backend system for validation. Before recording the vote, the system invokes the smart contract, which enforces predefined election rules such as one-vote-per-voter and candidate validity checks. Once validated, the transaction is digitally signed and transmitted to the blockchain network.

Upon successful blockchain confirmation, the vote is permanently stored on the distributed ledger. The system then updates the voter's status in the database to indicate that the vote has been cast, preventing further voting attempts. Finally, the system displays a confirmation message indicating that the vote has been successfully recorded. If the optional verification feature is enabled, the voter may receive a verification code to independently confirm that their vote exists on the blockchain. The activity concludes once confirmation is successfully displayed, completing the secure voting cycle.

Key Control Points in the Activity Diagram

- Decision node for voter ID verification
- Decision node for face validation
- Smart contract validation before blockchain storage
- Blockchain confirmation before updating voter status

The Activity Diagram clearly demonstrates the complete operational flow of the proposed voting system. It shows how biometric authentication, smart contract validation, and blockchain storage are integrated into a structured workflow. By enforcing validation at multiple decision points, the system ensures security, transparency, and prevention of duplicate or unauthorized voting, thereby maintaining the integrity of the election process. In addition, the diagram highlights the sequential dependency between identity verification, vote casting, and blockchain confirmation, ensuring that no step is executed without proper validation. It also emphasizes the coordinated interaction between off-chain components (database and face verification) and on-chain mechanisms (smart contracts and blockchain ledger), providing a balanced architecture that enhances reliability and trust in the overall voting system.

5. IMPLEMENTATION AND TESTING

5.1 IMPLEMENTATION APPROACH

The proposed blockchain-based secure e-voting system is implemented using a modular architecture to ensure scalability, security, and maintainability. The system integrates facial recognition for voter authentication and blockchain technology for secure vote storage. The implementation consists of several modules including voter registration, face verification, voting interface, blockchain vote recording, duplicate vote prevention, transaction verification, and database management. Each module performs a specific function to ensure the integrity and transparency of the voting process.

Module 1: Voter Registration Module

The voter registration module is responsible for registering eligible voters into the system. During registration, the administrator enters voter details and captures the voter's face image which is stored in the system database. The stored facial image acts as a reference for future authentication during voting.

Steps involved in voter registration:

- The administrator enters voter details such as voter ID and constituency.
- The system captures or uploads the voter's facial image.
- The image is stored in the database and used for face verification.
- The voter record is saved in the database with voting status initially set to false

This module ensures that only authorized voters are allowed to participate in the election.

Module 2: Face Verification Module

The face verification module authenticates voters using facial recognition technology before allowing them to vote. The system compares the captured face during voting with the stored image from the database.

Steps involved in face verification:

- The voter enters their voter ID in the polling interface.
- The system retrieves the stored face image corresponding to the voter ID.
- The webcam captures the current image of the voter.
- Facial features are extracted using a face recognition algorithm.
- The captured face is compared with the stored face image.
- If the similarity score exceeds the threshold, the voter is authenticated.

This process ensures that only the registered voter can cast the vote.

Module 3: Voting Interface Module

The voting interface provides a simple and user-friendly interface for voters to cast their votes. After successful authentication through face verification, the voter is presented with the list of candidates.

Steps involved in the voting interface:

- Display list of candidates along with party symbols.

- Allow the voter to select a candidate.
- Confirm the voter's selection before submitting the vote.
- Send the vote data to the backend server for blockchain processing.

Module 4: Blockchain Vote Recording Module

This module is responsible for securely storing votes on the blockchain using smart contracts. A smart contract deployed on the Ethereum blockchain records each vote and prevents duplicate voting.

Steps involved in vote recording:

- The backend server generates a transaction request.
- The smart contract function `castVote()` is invoked.
- The voter ID acts as a unique token to identify voters.
- The smart contract checks whether the voter has already voted.
- If the voter has not voted previously, the vote is recorded.
- The transaction is added to a blockchain block and mined.

The blockchain ensures that votes cannot be modified or deleted once recorded.

Module 5: Duplicate Vote Prevention Module

To prevent multiple votes from the same voter, the system uses a mapping structure in the smart contract. Each voter ID is stored as a unique key in the blockchain mapping.

Process for duplicate vote prevention:

- When a vote is submitted, the smart contract checks the `hasVoted` mapping.
- If the voter ID already exists in the mapping, the transaction is rejected.
- If the voter ID is not found, the vote is recorded and marked as used.

This mechanism ensures that even if the database is manipulated, duplicate voting is not possible because the blockchain acts as the final source of truth.

Module 6: Blockchain Transaction Verification

After a vote is successfully recorded on the blockchain, the system generates a transaction hash. The transaction hash serves as proof that the vote has been recorded.

Steps involved:

- The blockchain returns a transaction receipt after vote submission.
- The system extracts the transaction hash from the receipt.
- The transaction ID is displayed to the voter.
- The transaction can be verified in the blockchain explorer or Ganache interface.

This ensures transparency and verifiability of the voting process.

Module 7: Database Management Module

The database stores voter details, facial images, and voter status. However, the blockchain maintains the final record of votes.

Database responsibilities include:

- Storing registered voter details.

- Storing facial images used for verification.
- Tracking whether a voter has participated in the election.

Even if the database is modified or deleted, duplicate voting is prevented by the blockchain smart contract.

5.2 FUNCTIONAL AND ALGORITHMIC IMPLEMENTATION

The proposed blockchain-based electronic voting system integrates face verification, voter authentication, and blockchain technology to ensure secure and transparent voting. The system consists of multiple modules including voter registration, face verification, vote casting, and blockchain vote storage. The system architecture ensures that votes are stored securely on the blockchain and duplicate voting is prevented using smart contracts.

1.Voter Registration (Off Chain Registration)

In the proposed system, voter registration is handled through the backend and database instead of directly through the smart contract. The administrator registers voters by storing their voter ID and facial image in the system database.

During registration, the system stores:

- Voter ID
- Constituency
- Facial Image

This stored information is later used for face verification during the voting process.

Code Snippet:

```
app.post('/registerVoter', async (req, res) => {

  const { voterId, constituency } = req.body;

  db.query(
    "INSERT INTO voters (voterId, constituency, hasVoted) VALUES (?, ?, false)",
    [voterId, constituency],
    (err) => {
      if (err) {
        return res.json({ error: "Registration failed" });
      }

      res.json({ success: "Voter registered successfully" });
    }
  );
});
```

This registration ensures that only authorized voters can participate in the election

2. Cast Vote (castVote)

The castVote function is the core function of the smart contract. It is used to record a vote on the blockchain. When a voter selects a candidate, the backend server invokes this smart contract function. The vote details are then stored permanently in the blockchain ledger. Before recording the vote, the smart contract checks whether the voter has already voted.

Code Snippet:

```
function castVote(
    string memory _voterId,
    uint _candidateId,
    string memory _constituency
) public {

    require(!hasVoted[_voterId], "Voter already voted");

    votes.push(Vote({
        voterId: _voterId,
        candidateId: _candidateId,
        constituency: _constituency,
        timestamp: block.timestamp
    }));

    hasVoted[_voterId] = true;
}
```

This function ensures that every vote is securely recorded on the blockchain.

3. Duplicate Vote Prevention

To prevent duplicate voting, the smart contract uses a mapping data structure that tracks whether a voter has already cast a vote. Each voter ID is stored in the mapping after the vote is recorded. If the same voter attempts to vote again, the transaction is rejected automatically.

Code Snippet:

```
mapping(string => bool) public hasVoted;
```

When a voter casts a vote successfully, the voter ID is marked as true in the mapping.

If the voter tries to vote again, the smart contract throws an error:

Voter already voted

4. Vote Storage on Blockchain

Once the vote is validated, the vote details are stored in a Vote structure inside the smart contract. Each vote contains the following fields:

- Voter ID
- Candidate ID
- Constituency
- Timestamp

Code Snippet:

```

struct Vote {
    string voterId;
    uint candidateId;
    string constituency;
    uint timestamp;
}
Vote[] public votes;

```

Every vote is added to the votes array, which stores all recorded votes in the blockchain ledger. Because blockchain data is immutable, the stored votes cannot be modified or deleted after recording.

5. Face Verification Before Voting

Before allowing the voter to access the voting interface, the system performs face verification. The system captures the voter's face using a webcam and compares it with the stored facial image in the database. If the face matches, the voter is allowed to proceed to the candidate selection screen. Otherwise, the voter is rejected.

Code Snippet:

```

app.post('/verifyFace', async (req, res) => {

    const { voterId } = req.body;

    db.query(
        "SELECT * FROM voters WHERE voterId = ?",
        [voterId],
        (err, result) => {

            if (result.length === 0) {
                return res.json({ error: "Voter not found" });
            }

            res.json({ success: "Face verified successfully" });

        }
    );
});

```

This process ensures that only legitimate voters can participate in the election.

6. Blockchain Transaction Verification

After a vote is successfully recorded, the blockchain generates a transaction hash. This transaction hash acts as proof that the vote has been recorded. The backend retrieves this transaction hash and displays it to the voter.

Code Snippet:

```
app.post('/vote', async (req, res) => {  
  
  const { voterId, candidateId } = req.body;  
  
  const tx = await contract.castVote(  
    voterId,  
    parseInt(candidateId),  
    "HYD-01"  
  );  
  
  const receipt = await tx.wait();  
  
  res.json({  
    success: "Vote recorded on Blockchain",  
    txid: receipt.hash  
  });  
});
```

The transaction ID allows the voter to verify their vote on the blockchain network.

5.3 INTEGRATION DETAILS

The backend of the proposed system is implemented using Node.js and Express.js, which act as an interface between the frontend voting application and the blockchain network. The backend is responsible for processing user requests, validating voter information, and invoking smart contract functions to store votes securely on the blockchain.

The backend uses Ethers.js to communicate with the Ethereum blockchain network deployed locally using Ganache. The server receives requests from the frontend interface, verifies the input data, and submits transactions to the smart contract. Each API endpoint performs a specific operation such as voter registration, face verification, vote submission, and blockchain transaction verification. Once the transaction is successfully processed, the backend sends the response back to the frontend interface.

1. Voter Registration API

This API allows the administrator to register a new voter in the system. The voter's information such as voter ID and constituency is stored in the database. The voter's face image is also stored for future authentication.

Code Snippet:

```
app.post('/registerVoter', async (req, res) => {  
  
  const { voterId, constituency } = req.body;
```

```

db.query(
  "INSERT INTO voters (voterId, constituency, hasVoted) VALUES (?, ?, false)",
  [voterId, constituency],
  (err) => {
    if (err) {
      return res.json({ error: "Registration failed" });
    }

    res.json({ success: "Voter registered successfully" });
  }
);
});

```

This ensures that only registered voters can participate in the election.

2. Face Verification API

This API verifies the voter's identity before allowing access to the voting interface. The captured face image is compared with the stored face image in the database.

Code Snippet:

```

app.post('/verifyFace', async (req, res) => {

  const { voterId } = req.body;

  db.query(
    "SELECT * FROM voters WHERE voterId = ?",
    [voterId],
    (err, result) => {

      if (result.length === 0) {
        return res.json({ error: "Voter not found" });
      }

      res.json({ success: "Face verified successfully" });

    }
  );
});

```

If the face verification fails, the voter is rejected and cannot proceed to the voting interface.

3. Cast Vote API

This API is responsible for submitting the voter's vote to the blockchain. The vote is recorded using the smart contract function.

Code Snippet:

```
app.post('/vote', async (req, res) => {

  const { voterId, candidateId } = req.body;

  if (!voterId || !candidateId) {
    return res.json({ error: "Voter ID and Candidate ID required" });
  }

  try {

    const tx = await contract.castVote(
      voterId,
      parseInt(candidateId),
      "HYD-01"
    );

    const receipt = await tx.wait();

    db.query(
      "UPDATE voters SET hasVoted = true WHERE voterId = ?",
      [voterId]
    );

    res.json({
      success: "Vote recorded on Blockchain",
      txid: receipt.hash
    });

  } catch (error) {

    res.json({
      error: "Blockchain transaction failed"
    });

  }

});
```

Once the vote is recorded, the blockchain generates a transaction hash, which acts as proof that the vote has been successfully stored.

4. Blockchain Transaction Verification API

This API allows the system to verify whether the vote has been successfully recorded on the blockchain using the transaction hash.

Code Snippet:

```
app.get('/verifyVote/:txid', async (req, res) => {  
  
  const txid = req.params.txid;  
  
  try {  
  
    const receipt = await provider.getTransactionReceipt(txid);  
  
    if (receipt) {  
      res.json({ success: "Vote verified on blockchain" });  
    } else {  
      res.json({ error: "Transaction not found" });  
    }  
  
  } catch (error) {  
  
    res.json({ error: "Verification failed" });  
  }  
});
```

This verification ensures transparency and trust in the voting system.

5.4 TRANSACTION FLOW

1. The transaction flow describes the process through which a vote is securely recorded and verified on the blockchain network.
2. The process begins when the voter enters their voter ID in the voting interface. The system retrieves the voter's stored information and initiates the face verification process.
3. If the face matches the stored image, the voter is authenticated and allowed to access the candidate selection interface. If the face does not match, the voter is rejected and the voting process stops.
4. After authentication, the system displays the list of candidates. The voter selects a candidate and submits the vote. The backend server receives the request and invokes the smart contract function to record the vote on the blockchain.
5. Before storing the vote, the smart contract checks whether the voter has already voted. If the voter has previously voted, the transaction is rejected to prevent duplicate voting.
6. If the voter has not voted before, the vote is recorded on the blockchain. The blockchain network processes the transaction and generates a unique transaction hash.

7. The system then updates the voter status in the database and displays the transaction ID to the voter. This transaction ID can later be used to verify that the vote has been successfully recorded on the blockchain.
8. This workflow ensures secure voting, transparency, and prevention of duplicate voting in the proposed blockchain-based electronic voting system.

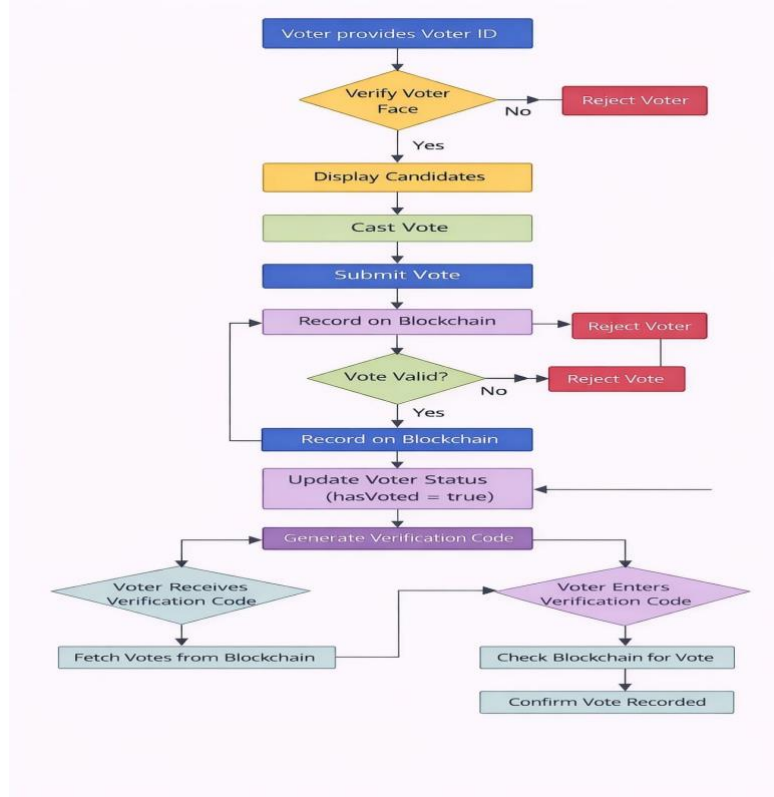


Fig. 5.1 – System Work Flow

5.5 TESTING

The testing phase is conducted to evaluate the functionality, performance, and reliability of the proposed Blockchain-Based Secure E-Voting System. This phase ensures that all modules of the system operate correctly and meet the intended requirements. Various system components such as voter verification, face authentication, vote casting, and blockchain transaction processing are tested to confirm that the system behaves as expected under different conditions.

During testing, different scenarios are executed to verify important functionalities including voter eligibility verification, successful face recognition, prevention of duplicate voting, and secure vote storage on the blockchain. Each test case compares the expected output with the actual output generated by the system. This process helps identify potential issues and ensures that the voting system operates securely, accurately, and transparently before deployment.

TEST CASES

The test cases are designed to verify the functionality, correctness, and reliability of the proposed Blockchain-Based Secure E-Voting System. Each test case evaluates whether a specific feature of the system behaves as expected under predefined conditions and inputs.

The testing process includes validating important functionalities such as voter verification, face authentication, vote casting, duplicate vote prevention, and vote verification through blockchain transactions. Each test case compares the expected output with the actual output produced by the system to ensure that the system performs correctly under different scenarios.

Table. 5.1 – Test Cases

S.No.	Test Scenario	Input Data	Expected Output	Actual Output	Status
1	Verify Voter	voterId: V101	Voter eligibility verified	Voter verified successfully	Pass
2	Face Authentication	Correct voter face	Face matched successfully	Face matched and access granted	Pass
3	Face Authentication Failure	Different face	Face mismatch detected	Verification failed and returned back	Pass
4	Display Candidates	Verified voter	Candidate list displayed	Candidate list displayed	Pass
5	Cast Vote	voterId: V101, candidateId: C1	Vote recorded on blockchain	Vote recorded successfully	Pass
6	Prevent Duplicate Vote	voterId: V101 voting again	Duplicate vote rejected	Error message displayed	Pass
7.	Generate Transaction ID	Vote submission	Transaction ID generated	Transaction ID displayed	Pass

6. RESULT ANALYSIS

The results obtained from the implementation of the proposed Blockchain-Based Secure E-Voting System demonstrate that the system performs effectively in delivering a secure, transparent, and reliable voting process. The voter authentication module based on face verification accurately identifies eligible voters and ensures that only authorized individuals are allowed to access the voting interface. This mechanism significantly reduces the chances of impersonation and unauthorized access, thereby improving the overall security of the voting process. After successful authentication, the system allows the voter to cast their vote through a simple and user-friendly interface. The vote is then securely transmitted to the blockchain network, where it is validated and recorded using smart contracts. The smart contract logic strictly enforces voting rules and ensures that each voter can cast only one vote, thereby effectively preventing duplicate voting attempts.

All recorded vote transactions are stored in the blockchain ledger, ensuring immutability and maintaining the integrity of the voting data. Since blockchain technology provides a decentralized and tamper-proof environment, the recorded votes cannot be modified, deleted, or manipulated after submission. This feature enhances transparency and builds trust in the entire voting system. In addition, the system generates a unique transaction ID after a vote is successfully recorded, allowing voters to verify that their vote has been included in the blockchain without revealing their identity. The integration between the frontend interface, backend APIs, and blockchain network works seamlessly, enabling efficient communication and smooth execution of voting operations. The system also provides appropriate confirmation messages and error notifications, ensuring clarity for users during the voting process. The overall performance of the system indicates minimal delay in processing vote transactions while maintaining strong security and reliability. These results clearly demonstrate that the proposed system successfully fulfills its objectives of secure voter authentication, transparent vote recording, prevention of duplicate voting, and trustworthy verification through blockchain technology, making it a suitable and efficient approach for modern electronic voting systems.

7. CONCLUSION & FUTURE SCOPE

7.1 CONCLUSION

The proposed Blockchain-Based Secure E-Voting System successfully demonstrates how blockchain technology can be utilized to enhance transparency, security, and reliability in the digital voting process. The system enables secure voter authentication using face verification and ensures that each vote is recorded as a transaction on the blockchain network. By leveraging smart contracts and decentralized blockchain architecture, the system guarantees that each vote is validated before being stored, eliminating the possibility of vote tampering or unauthorized modifications. This decentralized approach removes dependence on a central authority and ensures a fair and trustworthy voting process.

The implementation effectively prevents issues such as duplicate voting and identity fraud by combining facial authentication with blockchain-based transaction validation. Each vote recorded on the blockchain remains immutable and verifiable through a unique transaction ID, allowing voters to confirm that their vote has been successfully recorded without revealing their identity. The integration of a user-friendly web interface, backend APIs, and blockchain infrastructure ensures smooth system operation and accessibility. Overall, the system achieves its objective of providing a secure, transparent, and efficient digital voting platform capable of improving trust and integrity in modern electoral systems.

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7.2 FUTURE SCOPE

The proposed system can be further enhanced by integrating advanced technologies and additional features to improve its functionality and real-world applicability. One possible improvement is deploying the blockchain network on a public test network such as Sepolia Testnet instead of the local Ganache environment. This would allow the system to operate on a more realistic blockchain infrastructure and test transactions under real-world conditions. In addition, integrating biometric identity verification mechanisms such as Aadhaar-based authentication could strengthen voter verification and reduce the possibility of impersonation during the voting process.

Another enhancement is the development of a mobile application for the voting platform to improve accessibility and user convenience. A mobile-based system would allow voters to interact with the platform more easily while maintaining secure authentication and smooth user experience. It would also enable voters to participate in the voting process from different locations while ensuring proper security mechanisms are maintained. The scalability of the system can also be improved by deploying it across a distributed blockchain network with multiple nodes managed by different authorities. This would increase reliability, system availability, and fault tolerance during elections. These improvements can help transform the system into a more robust, scalable, and secure blockchain-based voting platform suitable for practical implementation in real-world voting scenarios.

8. REFERENCES

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APPENDIX I

SAMPLE CODE

Voting.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract Voting {

    struct Vote {
        string voterId;
        uint candidateId;
        string constituency;
        uint timestamp;
    }

    mapping(string => bool) public hasVoted;
    Vote[] public votes;

    function castVote(
        string memory _voterId,
        uint _candidateId,
        string memory _constituency
    ) public {

        require(!hasVoted[_voterId], "Voter already voted");

        votes.push(Vote({
            voterId: _voterId,
            candidateId: _candidateId,
            constituency: _constituency,
            timestamp: block.timestamp
        }));

        hasVoted[_voterId] = true;
    }

    function getVotesCount() public view returns(uint){
        return votes.length;
    }
}
```

```
}
```

server.js

```
app.post("/vote", async (req,res)=>{

const {voterId,candidateId} = req.body;

try{

const tx = await contract.methods
.castVote(voterId,candidateId,"Constituency1")
.send({from:account,gas:300000});

await db.query(
"UPDATE voters SET hasVoted = true WHERE voterId = ?",
[voterId]
);

res.json({
success:"Vote recorded on blockchain",
txId:tx.transactionHash
});

}
catch(err){

res.json({error:"Voting failed"});

}

});
```

voter.html

```
async function scanFace(){

const detection = await faceapi
.detectSingleFace(video,new faceapi.TinyFaceDetectorOptions())
.withFaceLandmarks()
.withFaceDescriptor();

if(!detection){
```

```

document.getElementById("faceMessage").innerText="No face detected";
return;
}

const distance = faceapi.euclideanDistance(
detection.descriptor,
storedDescriptor
);

if(distance < 0.50){

document.getElementById("faceMessage").innerText="Face Matched";
loadCandidates();

}
else{

document.getElementById("faceMessage").innerText="Face Mismatch";

}

}

```

Database Tables

-- Voters Table

```

CREATE TABLE voters (
  voterId VARCHAR(20) PRIMARY KEY,
  imagePath VARCHAR(255),
  hasVoted BOOLEAN DEFAULT FALSE
);

```

-- Candidates Table

```

CREATE TABLE candidates (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  partySymbol VARCHAR(255)
);

```

APPENDIX II

SCREENSHOTS

 Election Commission Admin Panel

3
Total Voters

4
Candidates

0
Votes Cast

Voters List

Voter ID	Has Voted	Action
v101	No	Delete
v102	No	Delete
v103	No	Delete

Candidates List

ID	Symbol	Name	Action
1		Candidate A	Delete
2		Candidate B	Delete
3		Candidate C	Delete
4		Candidate D	Delete

Polling Booth

Officer Verification

Verify

Eligible to Vote 



Secure Blockchain Voting System

Polling Booth

Face Authentication

Live Camera

Captured Face



No face detected. Adjust position...

Select Candidate



Candidate A

Vote



Candidate B

Vote



Candidate C

Vote



Candidate D

Vote

☒ Vote Recorded Successfully

Transaction ID:

0x8bbe32bb7b174a18364269135e656f2ce2ee448db27f6eb243f3a1a61f03f35c

Thank You For Voting

Voters List Refresh ↺

Voter ID	Has Voted	Action
v100	Yes	Delete
v101	Yes	Delete

Polling Booth

Officer Verification

Verify

Already voted ❌



Candidate A

Vote



Candidate B

Vote



Candidate C

Vote



Candidate D

Vote

Blockchain transaction failed ❌